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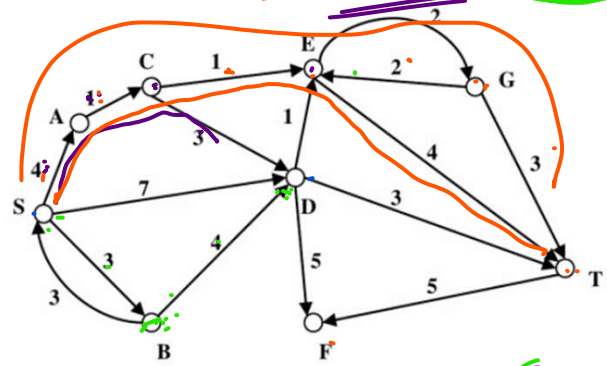
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GATE Questions on Dijkstra's Algorithm

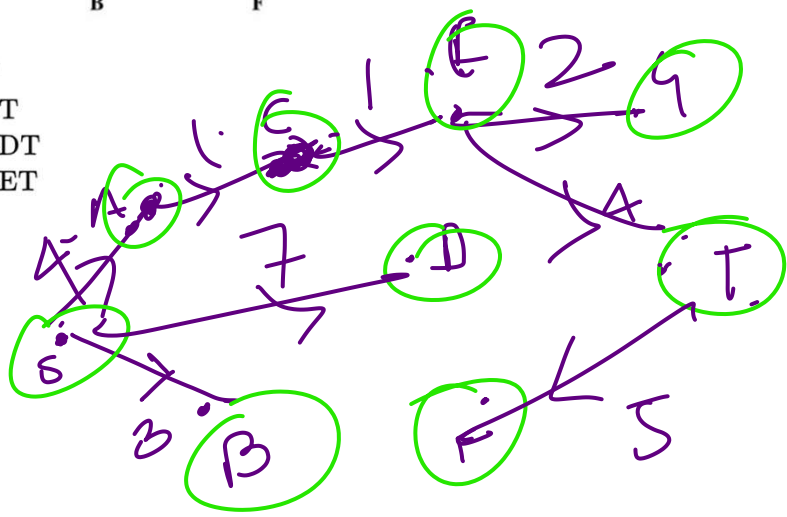
- Dijkstra is a greedy technique

gate 2012

Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T . Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.



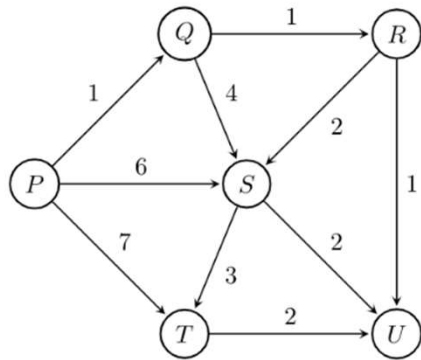
- A. SDT
- B. SBDT
- C. SACDT
- D. SACET



	S	A	B	C	D	E	F	G	T
S	0	∞	∞	∞	∞	∞	∞	∞	∞
B		4	5	∞	7	∞	∞	∞	∞
A		4	5	∞	7	∞	∞	∞	∞
C				1	7	∞	∞	∞	∞
E					7	1	∞	∞	∞
G							∞	2	4
T									5
F							7		
D									

gate 2007

Suppose we run Dijkstra's single source shortest path algorithm on the following edge-weighted directed graph with vertex P as the source.



In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized?

- A. P, Q, R, S, T, U
- B. P, Q, R, U, S, T
- C. P, Q, R, U, T, S
- D. P, Q, T, R, U, S

Home work

I need

